Multiple Regression II and Indicator Variables — Tips

Partial tests, multicollinearity, polynomial and qualitative predictors

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https://github.com/HelloJJJie/Assignment-Forecast-and-Analysis-of-Regression.git

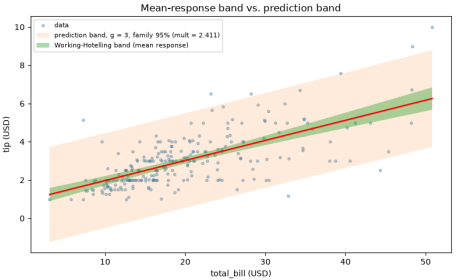
$n = 244$ restaurant bills — which predictors of the tip are *needed*, what happens when they overlap, and do groups differ?

1. Recap: simple linear regression (weeks 1–3)

Model. $\hat{Y} = 0.920 + 0.105\,X$ (X = bill, Y = tip, $n = 244$).

b_1	0.1050	R^2	0.457
$s\{b_1\}$	0.0074	r	0.676
95% CI	[0.0905, 0.1195]	s	1.02
t^*	14.26	F^*	203.4

- **Diagnostics:** constant variance rejected (Brown–Forsythe $P = 1.4 \times 10^{-7}$, Breusch–Pagan $P = 4 \times 10^{-21}$); normality rejected ($r = 0.982$ vs. 0.994); 5 cases with $|e^*| > 3$.
- **Simultaneous inference:** a family 95% interval for β_0, β_1 needs multiplier 2.255 (Bonferroni), not 1.970; the whole-line band uses $W = 2.463$.



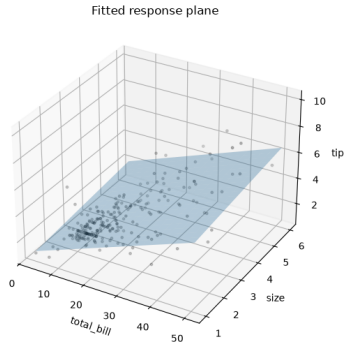
Prediction band ($g = 3$) vs. Working–Hotelling band: the new-tip band is about 15× wider at $\bar{X}$.

2. Recap: multiple regression and the hat matrix (week 4)

Model. $\hat{Y} = 0.669 + 0.0927X_1 + 0.1926X_2$ (X_2 = party size).

	b	$s\{b\}$	t^*
total bill	0.0927	0.0091	10.17
size	0.1926	0.0853	2.26

- **Fit:** $F^* = 105.95$, $R^2 = 0.468$, $R^2_a = 0.463$ — only 0.011 above the bill alone.
- **Hat matrix:** $\mathbf{H}$ symmetric, idempotent, $\operatorname{tr}\mathbf{H} = p$; 22 of 244 cases have $h_{ii} > 2p/n$.
- **Influence:** row 170 (bill \$50.81): leverage 0.071, $r = 4.14$, Cook's $D = 0.437$; dropping the top three cases moves the coefficients 12–19%.



Each coefficient holds the other predictor fixed. Variance is still not constant (Breusch–Pagan $\chi^2 = 92.2$).

3. Extra sums of squares and partial F tests

Y = tip, X_1 = bill, X_2 = size, X_3 = bill per person (X_1/X_2).

$SSR(X_2 \mid X_1) = SSE(X_1) - SSE(X_1, X_2)$: the extra error removed by adding X_2. **The order matters:**

order X_1, X_2, X_3		order X_3, X_2, X_1	
$SSR(X_1)$	212.42	$SSR(X_3)$	56.14
$SSR(X_2 \mid X_1)$	5.24	$SSR(X_2 \mid X_3)$	145.32
$SSR(X_3 \mid X_1, X_2)$	1.30	$SSR(X_1 \mid X_3, X_2)$	17.50
sum	218.96	sum	218.96

Both orders add up to the same $SSR = 218.96$ of $SSTO = 465.21$.

Partial F, one coefficient ($F^* = (t^*)^2$; $F(0.95; 1, 240) = 3.88$):

	F^*	t^*	P
total bill	17.05	+4.13	5.0×10^{-5}
size	0.03	−0.16	0.87
per person	1.27	−1.13	0.26

A pair at once ($F(0.95; 2, 240) = 3.03$):

Dropped together	F^*	P
size + per person	3.19	0.043
bill + per person	52.4	1.3×10^{-19}
bill + size	79.3	3.6×10^{-27}

Given the others, only the bill is needed — yet size and per-person *together* are borderline significant.

4. Partial R^2, standardization, multicollinearity

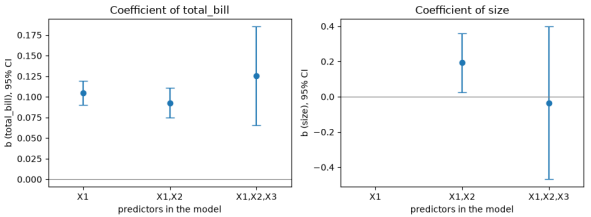
Partial R^2 (share of what the others leave that a predictor explains):

	R^2 partial	partial r
$R^2_{Y2\|1}$	0.0207	$+0.144$
$R^2_{Y3\|12}$	0.0053	-0.073
$R^2_{Y1\|23}$	0.0663	$+0.258$

Correlation transform ($\mathbf{r}_{XX}\mathbf{b}^* = \mathbf{r}_{YX}$):

	b_k	s_k	b^*_k
total bill	0.1253	8.90	$+0.806$
size	-0.0356	0.95	-0.024
per person	-0.0850	2.91	-0.179

Condition number 49,234 for $X'X$ vs. 67.8 for $\mathbf{r}_{XX}$.



Unstable and imprecise: the size coefficient flips from $+0.193$ to -0.036 when per person (derived: X_1/X_2) enters; $s\{b_1\}$ grows $4.1\times$.

VIF $= 17.3,\ 10.4,\ 11.4$ (1.56 without per person). Still fine: R^2 0.468 $\to$ 0.471, prediction at ($20, party of 3) 3.10 $\to$ 3.11.

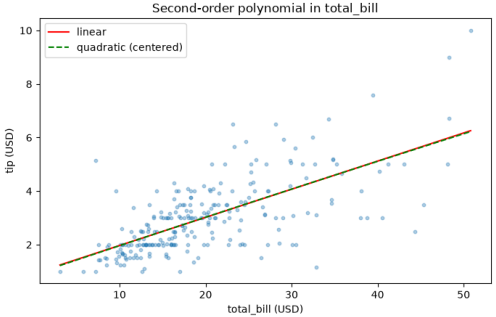
5. Polynomial regression and centering

Second-order model with the centered predictor $x = X - \bar{X}$:

$$Y = \beta_0 + \beta_1 x + \beta_{11} x^2 + \varepsilon$$

Still linear in the β's, so ordinary least squares applies.

- **Why center:** X and X^2 are strongly correlated over a positive range; centering makes $X'X$ better conditioned.
- **Back to the original scale:** $b_1 = b_1' - 2b_{11}'\bar{X}$, $b_0 = b_0' - b_1'\bar{X} + b_{11}'\bar{X}^2$ (checked against the raw fit).
- **Test the curvature:** $t^* = -0.09$, $P = 0.92$; R_a^2 $0.4544 \to 0.4521$. The fitted curve lies on the line.



	raw X	centered x
$\mathrm{corr}(X, X^2)$	0.971	0.633
cond. number of $X'X$	1.2×10^7	4.0×10^4

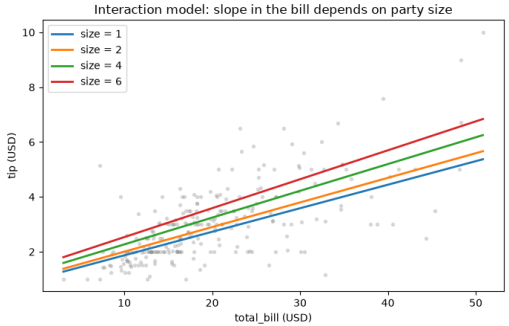
Not 0: the bill is right-skewed (skewness 1.13); it is exactly 0 only for symmetric X.

6. Interaction between two quantitative predictors

Interaction: $E\{Y\} = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2$; the slope in the bill is $\beta_1 + \beta_3 X_2$. Additive $\Rightarrow$ parallel, interacting $\Rightarrow$ not parallel.

party size	slope of the bill	95% CI
1	0.0858	[0.0545, 0.1172]
3	0.0937	[0.0753, 0.1121]
6	0.1055	[0.0545, 0.1565]

- $b_3 = +0.0039$, $t^* = 0.53$, $P = 0.60$: no evidence of interaction.
- **Full second-order model** (5 terms): none of x_1^2, x_2^2, $x_1 x_2$ is significant; together $F^* = 0.24$, $P = 0.87$; R_a^2 0.4635 $\rightarrow$ 0.4583.



The lines for party sizes $1, 2, 4, 6$ are nearly parallel.

So: for bill and size the additive first-order model is enough.

7. Qualitative predictors: indicator variables

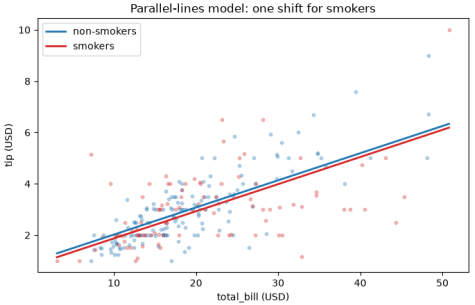
Coding. A class has no scale: use 0/1 indicators. $\mathbf{c} - \mathbf{1}$ **for** c **classes** — with c indicators plus the intercept, $\mathbf{X}'\mathbf{X}$ is singular. The all-zero class is the baseline.

Parallel lines ($Z = 1$ for smokers):

$$\hat{Y} = 0.963 + 0.1057 X_1 - 0.149\, Z$$

$b_2 = -0.149$, 95% CI $[-0.415, 0.117]$, $P = 0.27$: no vertical shift.

`day` has 4 classes: intercept + 4 indicators has rank 4 of 5.



baseline	Fri	Sat	Sun	Thur
Fri	–	−0.086	+0.075	−0.019
Sat	+0.086	–	+0.161	+0.067
Thur	+0.019	−0.067	+0.094	–

Each row: shift of the column's day relative to the baseline.

Same fit for every baseline (SSE 251.74).

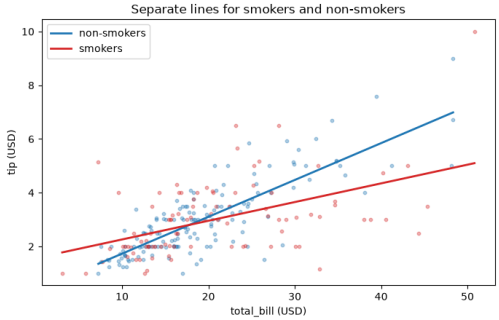
8. Comparing regression functions: smokers vs. non-smokers

$E\{Y\} = \beta_0 + \beta_1 X_1 + \beta_2 Z + \beta_3 X_1 Z$: slope β_1 for non-smokers, $\beta_1 + \beta_3$ for smokers.

Question	H_0	F^*	P
same slope?	$\beta_3 = 0$	22.7	3.3×10^{-6}
identical?	$\beta_2 = \beta_3 = 0$	12.0	1.1×10^{-5}
same height?	$\beta_2 = 0$	—	0.27^*

*in the parallel-lines model (t test).

- Slopes: 0.137 (non-smokers) vs. 0.070 (smokers).
- $\beta_2 = 0$ is read in the parallel model: $X_1 = 0$ is outside the data.
- $SSE(F) = 229.81$ equals the sum of the two separate regressions.



The smokers' line is flatter: tips rise more slowly with the bill.

Robust check: with heteroskedasticity-robust (HC3) errors the slope difference is still significant, $P = 0.008$ (ordinary: 3×10^{-6}).

9. Other qualitative predictors, and what the data say

More than two classes: $c - 1$ indicators, and their interactions with X_1 give a slope per class. Same two tests, $\alpha = 0.05$:

	equal slopes		identical	
	F^*	P	F^*	P
smoker (2)	22.7	3.3×10^{-6}	12.0	1.1×10^{-5}
sex (2)	0.30	0.58	0.17	0.85
time (2)	2.10	0.15	1.05	0.35
day (4)	3.66	0.013	2.00	0.066

df: $(1, 240)$, $(2, 240)$; for day $(3, 236)$, $(6, 236)$.

- **Smoker** is the only grouping that changes the line; sex and time show nothing; day is borderline.
- The tests assume constant variance, which fails here (week 3); the HC3 check on smoker supports it.

What the tips data say

- The bill explains 46% of the variance of the tip; party size adds about 1 point of R^2.
- No curvature and no bill $\times$ size interaction.
- A predictor built from the others (bill per person) adds no fit and breaks the coefficients (VIF > 10, size flips sign).
- Smokers' tips grow 0.070 per dollar of bill, non-smokers' 0.137.

Caveats. The variance grows with the bill and the residuals are not normal, so intervals for new tips are too narrow at high bills; a few high-leverage bills move the coefficients by 12–19%.